

S1Q: Low-Bit Quantization for System One Decision Models

Yuanming Chen

CYMCharming

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Abstract

Typed decision models map a state and structured questions directly to distributions over supplied choices, Boolean answers, or ordered score levels. Their deployment calls for quantization evaluations that measure decision and probability preservation in addition to storage. We present S1Q, an open post-training quantization framework that adapts activation-aware channel scaling and groupwise clipping to native decision interfaces, with an optional label-free categorical-Fisher sensitivity estimator. A held-out development objective selects among weight-only, selectively protected, and simulated activation-quantized profiles. We evaluate Kev-0.8B, Kev-4B, Kev-9B, NanoJev, and Laya, spanning hybrid decoder, decoder-only, and encoder architectures. On exploratory main cohorts, selected S1Q improves or ties matched round-to-nearest quantization; four of five paired 95% intervals include zero. A separate fixed-profile evaluation on 85 screened public JevBench tasks yields heterogeneous outcomes, with all five improvement intervals including zero. Estimated complete parameter storage falls by 48.4–70.6%, and measured persistent CUDA allocations fall by 48.2–69.4%. Packed execution still performs floating-point matrix multiplication, so these reductions do not establish integer-kernel acceleration. S1Q contributes native adapters, explicit sensitivity and selection procedures, pinned evaluation recipes, and a reproducible account of both gains and failures.

1 Introduction

Many model-mediated decisions have a finite output space: authorize or deny a request, choose one supplied action, or assign a discrete score. The System One model paradigm and Jev’s typed API motivate direct prediction of such distributions rather than an autoregressive answer trace [13]. Independent open implementations include Kev, NanoJev, and Laya [14, 15, 16]. We use *System One decision model* operationally for this interface, not as a claim that these models share a standardized architecture. The hosted TypeSafe Jev service is not an open checkpoint evaluated here.

Low-bit quantization is a practical way to reduce pa-

rameter storage. Strong language-model methods include GPTQ, AWQ, SmoothQuant, and OmniQuant [1, 2, 3, 4]. However, a deployment decision can change when the two leading candidate probabilities are close, even if an aggregate reconstruction error is small. Conversely, preserving an original model’s probabilities does not imply that those probabilities describe empirical truth well. A decision-focused study must distinguish accuracy, fidelity to the native model, label-based probability quality, and system resource use.

Architecture and interface details also matter. Kev combines a Qwen hybrid backbone, merged adapters, and a pointer head. NanoJev uses a Qwen3 backbone with dynamic candidate heads. Laya uses a ModernBERT encoder and typed heads. Candidate numbers vary by question, and a request can contain multiple questions. Generic causal-language-model evaluation or a rewritten prompt can silently change the task. We quantize through pinned native adapters and retain each implementation’s admitted question encoding and candidate order.

S1Q adapts established groupwise quantization, activation-aware scaling, and clipping search to these interfaces. Its optional Fisher variant estimates how intermediate perturbations affect the teacher’s categorical decision distribution. Random projections accommodate variable candidate sets without constructing a full propagated Fisher matrix. A finite development search chooses the entire profile rather than assuming that additional sensitivity weighting or lower activation precision always helps. We do not claim that scaling, clipping, Fisher sensitivity, or protected heads are individually new.

The empirical findings are mixed. Main-cohort accuracy differences against matched RTN are nonnegative, but most confidence intervals include zero. Source transfer includes losses, and an external authored cohort does not show consistent improvement. Raw main negative log-likelihood is worse than RTN for four of five models. The Fisher variant is selected for only one model. These observations motivate a conservative account of what decision-aware quantization preserves and what it can damage.

Our contributions are:

1. A native PTQ framework and cross-architecture study of five released decision checkpoints, with explicit quantized and retained module boundaries.

2. A specified adaptation combining input second moments, optional categorical-Fisher probes, groupwise scale/clipping search, and development-based decision preservation.
3. Paired accuracy, probability, transfer, and storage analyses, including an externally frozen authored cohort and disclosed negative results.
4. Pinned recipes, paired predictions, and packed linear artifacts where redistribution is permitted, with measured persistent-memory reductions separated from unimplemented integer-kernel acceleration.

Code and artifacts are available at github.com/CYMCharming/S1Q.

2 Related Work

Post-training quantization. GPTQ uses approximate second-order reconstruction and error compensation [1]. AWQ motivates activation-guided channel scaling [2], while SmoothQuant uses equivalent transformations to support weight-activation quantization [3]. OmniQuant learns clipping and equivalent transformations through reconstruction [4]. S1Q uses a small discrete grid and a factorized diagonal proxy; it is neither an official reproduction of these algorithms nor evidence of superiority to them. Rotation-based approaches, including QuaRot and SpinQuant, and W4A8 systems such as QQQ illustrate the separate roles of numerical conditioning and efficient kernels [5, 6, 7].

Sensitivity and uncertainty. Fisher-based quantization sensitivity has been studied by FIT [8]. GuidedQuant incorporates end-loss guidance [9]. C-PTQ uses Fisher-weighted channel sensitivity and is especially close to the motivation for S1Q’s optional row weights [10]. We treat Fisher weighting as established prior art; our implementation adapts teacher categorical probes and question normalization to native typed decisions. DPQ targets uncertainty through calibration-data selection [11]. S1Q’s development objective uses decision margins and probability drift, but does not implement DPQ’s data-selection procedure. Temperature scaling is a separate probability-calibration operation [12].

Quantized decision models. Public Laya INT4 ONNX artifacts, Laya MLX engineering experiments, Kev INT8 evaluations, and quantized Kev GGUF backbones precede this study [18, 19, 20, 21]. Thus S1Q is not the first quantization of this model class. Different revisions, tasks, temperatures, and runtimes prevent treating their reported values as controlled S1Q comparisons. Our emphasis is a common audited procedure with native interfaces, paired probability measurements, and explicit failure disclosure.

3 Problem and Evaluation Contract

A request contains a state and Q typed questions. Question q has a finite candidate set of size K_q and teacher logits $z_q \in \mathbb{R}^{K_q}$. A Boolean question uses the common ordering [false, true]; NanoJev’s native scalar Boolean logit z is represented as $[0, z]$. Choice insertion order is retained. Score levels are treated as categories for the Fisher calculation; ordinal expected-level error is reported separately.

Let θ denote the original checkpoint and $\hat{\theta}$ the quantized model. We seek low parameter storage while preserving native decision behavior. This is not a guarantee that the teacher is correct or calibrated. Task labels are used for final scoring and development selection, but are removed from inference inputs and from PTQ statistic collection.

We quantize supported backbone Linear weights. Embeddings, decision heads, normalization, biases, convolution, and recurrent-state arithmetic retain their loaded precision. Shared parameters crossing the boundary are rejected rather than silently changing a protected module. Kev adapters are merged before collecting statistics. Each compared method uses the same admitted records and candidate semantics; inputs are not truncated to rescue a failed quantized prediction.

4 S1Q Method

4.1 Input Second Moments

For a layer $y = Wx + b$, $W \in \mathbb{R}^{d_{\text{out}} \times d_{\text{in}}}$, collect the pristine teacher’s input activations on calibration requests. Flatten leading dimensions into rows and compute

$$m_j = \mathbb{E}_{\text{cal}}[x_j^2], \quad h_j = \frac{m_j}{\max(\text{mean}_k m_k, 10^{-30})}. \quad (1)$$

These are *uncentered* second moments, not a centered covariance. The collector includes padding rows emitted by the native forward. Normalization does not change candidate ordering within a layer.

4.2 Optional Categorical-Fisher Probes

At fixed $T > 0$, define

$$u_q = z_q/T, \quad p_q = \text{softmax}(u_q), \quad F_q = \text{diag}(p_q) - p_q p_q^\top. \quad (2)$$

F_q is the categorical Fisher in scaled-logit coordinates. It uses the teacher distribution, not a gold label or a task reward. For independent $\epsilon_q \sim \mathcal{N}(0, I_{K_q})$, set

$$a_q = \text{diag}(\sqrt{p_q}) \epsilon_q, \quad (3)$$

$$v_q = a_q - p_q \mathbf{1}^\top a_q, \quad (4)$$

$$\ell_{\text{probe}} = Q^{-1/2} \sum_q \text{stopgrad}(v_q)^\top u_q. \quad (5)$$

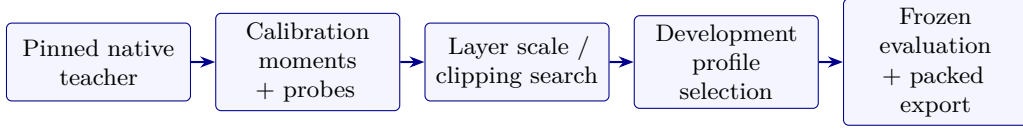


Figure 1: S1Q separates calibration statistics, local quantizer search, whole-model development selection, and fixed-profile evaluation. PTQ moments and probes are label-free; development selection and optional probability-temperature fitting use separate labeled partitions.

p_q and v_q are detached during differentiation. The scalar is a device for collecting sensitivity, not a training loss that is minimized.

Proposition 1 (Conditional probe covariance). *Fix a teacher, a calibration record, and a differentiable intermediate output y , vectorized across token rows and channels. Write $J_q = \partial u_q / \partial y$ and $g = \nabla_y \ell_{\text{probe}}$. Then*

$$\mathbb{E}[v_q] = 0, \quad \mathbb{E}[v_q v_q^\top] = F_q, \quad (6)$$

$$\mathbf{1}^\top v_q = 0, \quad \mathbb{E}[g] = 0, \quad (7)$$

$$\mathbb{E}[g g^\top] = Q^{-1} \sum_q J_q^\top F_q J_q. \quad (8)$$

The $Q^{-1/2}$ factor makes the expected squared gradient correspond to question-averaged sensitivity within a request. The common-shift identity removes sensitivity to adding the same constant to every candidate logit. It does not imply independence of token rows or independence of the questions’ representations.

For output channel i , accumulate squared gradients over records, probes, and flattened output rows:

$$\hat{f}_i = \frac{\sum_r \sum_{k=1}^P \sum_{t=1}^{n_r} (g_{r,k})_{ti}^2}{P \sum_r n_r}. \quad (9)$$

For a nonzero estimate, the implemented row weight is

$$r_i = \max \left(\frac{\hat{f}_i}{\max(\text{mean}_k \hat{f}_k, 10^{-30})}, 10^{-4} \right). \quad (10)$$

An all-zero estimate and the activation-only variant use $r_i = 1$. There is no renormalization after the floor. The normalized, floored weight is not an unbiased Fisher estimate; the covariance property applies to the raw probes and squared gradients. Parameters are frozen during collection, so backward passes do not update them. Failed records discard all partial statistics and do not trigger a silent activation-only fallback.

4.3 Equivalent Scaling and Groupwise Quantization

For exponent α , preliminary observed-channel scales are $\hat{s}_j = \max(\sqrt{h_j}, 10^{-8})^\alpha$. Normalize by the geometric midpoint of the observed minimum and maximum, then clamp to $[10^{-4}, 10^4]$. Unobserved channels receive one; $\alpha = 0$ sets every scale to one. With $S_\alpha = \text{diag}(s)$,

$$W'_\alpha = W S_\alpha, \quad x'_\alpha = S_\alpha^{-1} x. \quad (11)$$

This preserves $Wx + b$ before rounding in exact arithmetic.

Groups run along the input dimension independently in each output row, with default group size 128 and zero padding of a final incomplete group. For bit width b , clipping ratio c , and group g :

$$q_{\max} = 2^{b-1} - 1, \quad (12)$$

$$d_{ig} = \frac{c \max_{j \in g} |(W'_\alpha)_{ij}|}{q_{\max}}, \quad (13)$$

$$q_{ij} = \text{clip} \left(\text{round} \left(\frac{(W'_\alpha)_{ij}}{d_{ig}}, -q_{\max}, q_{\max} \right) \right). \quad (14)$$

All-zero groups use $d_{ig} = 1$. No affine zero point is stored. W4 uses $[-7, 7]$ and W8 uses $[-127, 127]$; the unused most-negative code is intentional. Clipping rescales a group’s maximum absolute range; it is not a percentile filter or removal of a fraction of weights. Define the reconstructed effective weight

$$\widehat{W}_{\alpha,c} = \mathcal{Q}_{b,c}(W S_\alpha) S_\alpha^{-1}. \quad (15)$$

4.4 Local Reconstruction Objective

Each layer independently minimizes

$$E_\ell(\alpha, c) = \sum_{i,j} r_i h_j ((\widehat{W}_{\alpha,c})_{ij} - W_{ij})^2 \quad (16)$$

over $\alpha \in \{0, 0.25, 0.5, 0.75\}$ and $c \in \{0.9, 0.95, 1\}$. Exact ties prefer smaller α , then larger c . The 12-point grid includes RTN at $(\alpha, c) = (0, 1)$. RTN performs no calibration-based search. The activation-only profile sets $r_i = 1$.

Equation (16) has a local motivation. For a small scaled-logit perturbation δu_q ,

$$\text{KL}(p_q \| \text{softmax}(u_q + \delta u_q)) = \frac{1}{2} \delta u_q^\top F_q \delta u_q + O(\|\delta u_q\|^3). \quad (17)$$

Local linearization gives $\delta u_q \approx J_q \delta y$. For a single layer with teacher inputs held fixed, $\delta y = \Delta W x$. Dropping output-channel, token-position, and input-channel cross terms, and factorizing activation magnitude from output sensitivity, motivates a term proportional to $\sum_{ij} \mathbb{E}[H_{ii}] \mathbb{E}[x_j^2] (\Delta W_{ij})^2$, where $H = Q^{-1} \sum_q J_q^\top F_q J_q$. The moment normalizations and sensitivity floor further modify this proxy. Cross-layer interactions and changes to downstream inputs are omitted. Thus the exact covariance in Proposition 1 does not make the diagonal reconstruction objective exact or guarantee task gains.

4.5 Protection and Activation Profiles

Optional protection ranks eligible layers by calibrated RTN relative error:

$$\rho_\ell = \frac{\sum_{ij} r_i h_j (\widehat{W}_{ij}^{\text{RTN}} - W_{ij})^2}{\sum_{ij} r_i h_j W_{ij}^2}. \quad (18)$$

A zero denominator yields zero. Preserve the largest $\lceil \eta L \rceil$ scores, breaking ties by module name; the tested η is 0.05. This budget counts layers, not parameters. The native protected boundary remains in effect even when $\eta = 0$.

Optional A4/A8 applies dynamic symmetric quantize–dequantize (QDQ) to each selected linear input row *after* compensation. One scale per row is its maximum absolute feature value divided by 7 or 127. Values are rounded, clamped, and immediately reconstructed in the original floating dtype. Other activations and recurrent states remain native. This is numerical simulation, not an integer activation runtime or KV-cache quantization.

4.6 Whole-Model Development Selection

Each profile starts from the pristine teacher and uses either uniform or Fisher output weights. Profiles are RTN, local, local with protection, local with A8/A4 QDQ, and, when enabled, Fisher with or without protection. There is no Fisher-plus-A4/A8 candidate in the current grid. RTN is a baseline, not a candidate for the selected S1Q label.

For development question n , let p_n and $\hat{p}_n$ be probabilities from raw teacher and quantized logits at temperature one. Let m_n be the difference of the teacher’s largest two probabilities:

$$w_n = 1 + \text{clip}((0.2 - m_n)/0.2, 0, 1), \quad (19)$$

$$\mathcal{J}_{\text{dev}} = \frac{\sum_n w_n \text{JS}(p_n, \hat{p}_n)}{\sum_n w_n} + \frac{0.1}{N} \sum_n H_n, \quad (20)$$

$$H_n = \mathbf{1}[\arg \max p_n = y_n, \arg \max \hat{p}_n \neq y_n]. \quad (21)$$

The weight lies in $[1, 2]$ and emphasizes near-boundary decisions. The harmful-flip term uses development labels; S1Q is not entirely label-free. Only exact objective ties are broken by estimated complete packed parameter bytes. The selection is persisted before evaluating quantized test cohorts. Reporting temperatures are fitted separately and do not enter Equation (20).

4.7 Storage and Reference Execution

W4 packs two signed nibbles into each byte. Artifacts contain integer payloads, FP32 group scales, and input-channel scales. Biases, retained native weights, architecture, heads, and tokenizer are obtained from the pinned native checkpoint. An artifact is a packed linear derivative, not a standalone compressed checkpoint.

The reversible accuracy path reconstructs floating weights and does not reduce their tensor dtype. The packed runtime instead replaces selected modules with

Algorithm 1: S1Q

1. Pin the teacher and native adapter; freeze candidate ordering, module boundary, admitted records, and data partitions.
2. Collect input moments; optionally collect categorical-Fisher output moments from the unchanged teacher.
3. For each profile, rank optional protected layers, then search (α, c) independently in remaining linears.
4. Evaluate compensated reconstructed weights and optional activation QDQ on development requests.
5. Select the S1Q profile minimizing $\mathcal{J}_{\text{dev}}$; persist its layer policy and parameters.
6. Fit separate reporting temperatures; evaluate frozen profiles on paired test, transfer, and external cohorts.
7. Pack integer codes and save scales, module identities, model pins, and hashes.

Figure 2: Calibration controls local quantizer parameters; development controls the whole-model profile. External data controls neither.

integer buffers and transiently dequantizes one matrix per call before floating Linear computation. It has no INT4/INT8 GEMM kernel. Loading first constructs the full native model; measured inference peaks exclude that initial allocation. Functional callers that directly access a reconstructed weight can bypass the A4/A8 input hook, so architecture coverage requires native parity checks.

5 Experimental Setup

Models and hardware. We evaluate three Kev sizes, NanoJev, and Laya at immutable source and checkpoint revisions (Appendix D). Kev experiments use an A100 80GB GPU; NanoJev and Laya use an A800 80GB GPU. Kev backbones are stored in BF16 with native FP32 head components. NanoJev and Laya native parameters are FP32 with BF16 autocast computation; a BF16 compute flag does not imply that their baseline parameter storage is BF16. Recorded package versions and a separately tested supported CPU profile are distinguished in Appendix E.

Quantizer budget. The main study fixes W4, group size 128, and the native head policy. It does not optimize bit width globally. All five completed expanded runs consider the optional Fisher profiles, using two probes per request and up to 32 accepted calibration requests at $T = 1$. Input statistics use the frozen calibration set. Local candidate reconstruction uses FP32, followed by execution in native compute dtypes. The protected profiles retain 5% of eligible layers. W8 support is available in code but is not an executed supplementary comparison in this snapshot.

Data. The shared screened Kev decision-v7 test has 711 requests, 914 questions, and 631 clusters; source-transfer-v4 has 630 requests/questions and 518 clusters. The shared quantizer, temperature, and development partitions have respectively 128/155, 128/155, and 256/311

requests/questions. Decision-v7 mixes public text tasks and deterministic rule scenarios and shares source families with Kev’s supervision. Transfer source families are held out from Kev supervision, not certified unseen by every other model.

NanoJev’s evaluated native-release cohorts contain *only shooting* states. Test has 1,023 decisions in 102 episode components, and OOD has 1,024 in 150 components. Every evaluated label is marked **reference_argmax_compatibility**: it is a supplied argmax action from an upstream visual reference policy. These are neither human judgments nor observed success probabilities. The converter copies explicitly supplied gold; it does not manufacture gold from S1Q or teacher predictions. The broader upstream package also contains other games, but those are not the final evaluated cohorts.

The independent external source is specifically **fstandhartinger/jevbench** [17]. A declared maximum of 10 candidates and 1,000 complete-request characters admits 85 of 183 public original-plus-hard tasks: 72 original and 13 hard, in 49 scenario clusters. Ninety-eight hard tasks are excluded for length. The retained questions comprise 40 Choice, 33 Boolean, and 12 Score tasks. Hard labels have model-assisted authorship and cross-review. No additional native adapter exclusions occur for these five external runs. This is a screened public authored cohort, not the full benchmark or a leaderboard result.

Split and development disclosure. Exact public requests and source groups are checked across partitions. NanoJev episodes linked by duplicate requests are joined into components; lower-priority overlapping components are removed whole before sampling. This does not establish absence of paraphrase duplicates or unknown pre-training contamination. Early Laya pilot test/transfer results informed the Fisher extension. Expanded main, transfer, and game OOD evaluations reuse overlapping inspected sources and remain exploratory. The external study freezes selection, artifacts, calibration policy, and data hashes before its first prediction and performs no external fitting or tuning.

Comparisons and metrics. We compare native checkpoints, matched RTN, an unprotected local ablation, and selected S1Q. RTN matches weight format, group size, quantized tensor identities, protection, and activation scheme. Kev-9B uses A8-matched RTN rather than ordinary W4 RTN. Official AWQ/GPTQ/OmniQuant experiments are absent; the local ablation must not be relabeled an official implementation. We report accuracy, NLL, multiclass Brier, 15-bin top-label ECE, JS drift, harmful flips, and score expected-level MAE. Paired percentile cluster bootstrap uses 2,000 resamples and seed 20261001. It is not source-stratified and has no multiple-comparison correction.

6 Results

6.1 Main Accuracy and Native Loss

Selected S1Q improves or ties matched RTN on the main cohorts (Table 1), but four of five difference intervals include zero. Kev-9B gains 2.19 percentage points, with an interval of [0.76, 3.71]. These inspected, expanded experiments are exploratory; this interval is not an independent confirmation of a general gain. Kev-0.8B and Laya lose 1.75 and 2.30 points against native checkpoints, respectively. Their descriptive native-difference intervals are entirely negative. W4 is therefore not lossless, and the data do not support uniform noninferiority to native inference.

6.2 Transfer and Native-Release OOD

Transfer results are heterogeneous (Table 2). Kev-4B gains 2.06 points over matched RTN, while Kev-0.8B loses 2.38 and Kev-9B loses 0.16. NanoJev’s game OOD gain is 0.88 points with an interval spanning zero. A smaller local surrogate or better main-cohort fidelity does not guarantee better accuracy under source or domain shift.

6.3 Frozen External Evaluation

On the screened JevBench cohort, selected-minus-RTN differences range from -3.53 to $+4.71$ points (Table 3, Figure 3). All five paired intervals include zero. Point estimates include gains, a tie, and losses; this small authored set does not establish consistent cross-model improvement. An interval crossing zero is also not evidence of equivalence.

NanoJev’s native general-domain accuracy is already 40.00%; S1Q yields 38.82%, a further -1.18 -point change. The low absolute native score is consistent with a transfer limitation before quantization, while the paired difference isolates an additional quantizer effect. We do not infer general-purpose decision ability from its specialized shooting-game evaluation.

6.4 Ablations and Probability Quality

Fisher is selected for only Kev-0.8B; the other four models choose local activation-aware scaling, with Kev-9B adding A8 QDQ. No selected profile retains the optional extra 5% of layers. The development grid does not establish a general Fisher benefit. Full development accuracy and objective values are reported in Appendix B.

A4 QDQ reduces development accuracy relative to native for every model. The drops are large for the Kev checkpoints and Laya, but much smaller for NanoJev. This motivates model-specific activation checks, not a successful universal W4A4 recipe.

Raw main NLL is worse than matched RTN for Kev-0.8B, Kev-4B, Kev-9B, and Laya. Accuracy or teacher

Model	Selected policy	n	Native	RTN	S1Q	Δ vs. RTN [95% CI]	Δ vs. native [95% CI]
Kev-0.8B	Fisher W4	914	82.93	80.63	81.18	+0.55[−1.07, +2.18]	−1.75[−3.06, −0.45]
Kev-4B	Local W4	914	85.45	84.79	84.79	+0.00[−1.36, +1.30]	−0.66[−1.76, +0.43]
Kev-9B	Local W4A8 [†]	914	86.98	84.35	86.54	+2.19[+0.76, +3.71]	−0.44[−1.44, +0.55]
NanoJev	Local W4	1023	79.86	79.47	80.45	+0.98[−0.11, +2.31]	+0.59[−0.30, +1.48]
Laya	Local W4	914	66.85	63.68	64.55	+0.88[−1.27, +3.10]	−2.30[−4.25, −0.23]

Table 1: Exploratory main-cohort accuracy (%). Differences and interval endpoints are percentage points. Kev/Laya use the shared 914-question cohort; NanoJev uses 1,023 reference-policy action targets. [†]A8 is simulated QDQ and is matched in RTN. Intervals describe paired differences, not absolute-accuracy uncertainty.

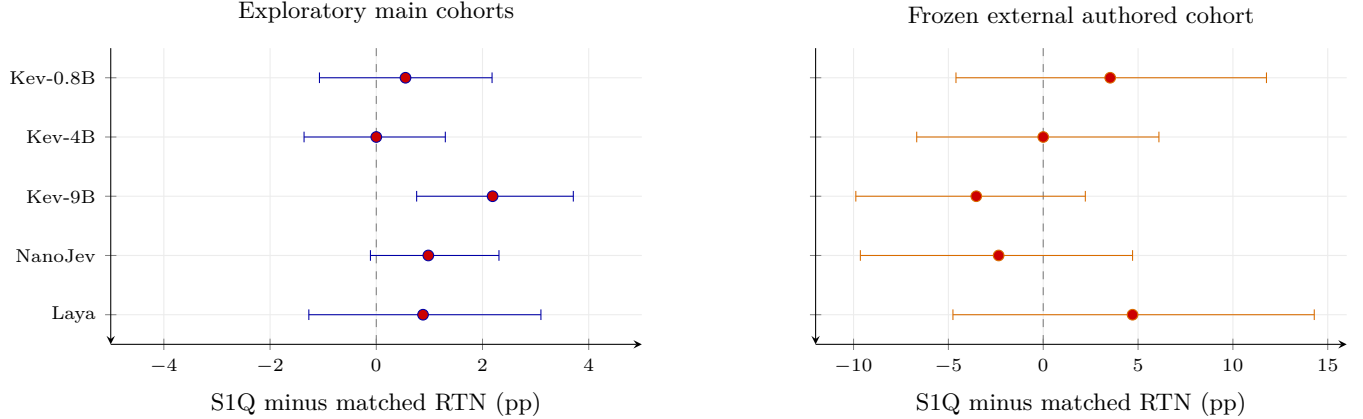


Figure 3: Paired 95% cluster-bootstrap intervals. The external panel uses a wider horizontal scale because the small cohort has greater uncertainty. Every external interval crosses zero; neither consistent improvement nor equivalence follows. Cohort definitions and profiles remain distinct.

fidelity can improve while label-based probability quality degrades. Temperature fitting is offered to every compared method on its separate labeled partition; it does not change argmax accuracy. Appendix C reports raw and temperature-scaled NLL, Brier, and ECE. None establishes real-world event calibration for authored or reference-policy targets.

6.5 Storage and Persistent Memory

Estimated complete parameter storage, including retained native components and quantizer auxiliaries, is 29.4–51.6% of native storage (Table 4). The denominator reflects actual native storage precision, not a uniform hypothetical FP16 model. Quantized parameter coverage varies from 66.09% to 87.15%. A small exported linear file should not be reported as the complete-model compression ratio.

Measured persistent PyTorch CUDA allocation decreases by 48.2–69.4% (Table 5). For Kev-9B it falls from 15,154.2 to 5,488.1 MiB, with a packed inference peak of 6,107.1 MiB. The initial full native checkpoint load is excluded from that peak, so these figures do not establish direct loading on a device of the packed persistent size.

Parity checks cover 41 scored decisions for each Kev model and Laya, and 32 for NanoJev. Zero JS and flips on these subsets do not establish bitwise equality or full-test packed validation. The benchmark records packed request

times but lacks paired dense latency measurements; we report no speedup ratio. All matrix multiplications remain floating point.

7 Limitations and Research Boundaries

S1Q combines established ingredients in a native decision framework. Close sensitivity-based prior work prevents treating Fisher weighting as a new general principle, and direct quantized Kev/Laya artifacts prevent a broad first-work claim. Official AWQ, GPTQ, OmniQuant, and close gradient-guided baselines have not been run under this contract. RTN and a local ablation are useful controls but cannot establish state-of-the-art quantization.

The study uses one reported calibration/probe seed and a restricted profile grid. Multiple seeds, calibration sizes, and probe counts are required to assess stability and estimator cost. The diagonal factorization ignores dependencies; padded token rows enter the statistic averages, and score-level ordering is absent from the categorical Fisher. Parameter selection targets teacher preservation with a small labeled harm term, not a formal deployment risk guarantee.

Main and transfer evaluations are exploratory after pilot inspection. The external set is small, authored, and heavily length-screened; unknown pretraining membership

Model	n	Native	RTN	S1Q	Δ vs. RTN [95% CI]
Kev-0.8B	630	68.25	68.25	65.87	-2.38[-5.38, +0.48]
Kev-4B	630	83.97	83.17	85.24	+2.06[+0.16, +4.10]
Kev-9B	630	85.24	84.44	84.29	-0.16[-1.79, +1.57]
Laya	630	66.03	65.87	66.83	+0.95[-1.93, +3.93]
NanoJev	1024	77.44	76.95	77.83	+0.88[-0.09, +1.86]

Table 2: Exploratory source transfer (Kev/Laya) and native-release game OOD (NanoJev). Accuracy is %; differences are pp. Transfer uses 518 clusters; NanoJev OOD uses 150 episode components. Source holdout is defined relative to Kev supervision.

Model	n	Native	RTN	S1Q	Δ vs. RTN [95% CI]
Kev-0.8B	85	76.47	72.94	76.47	+3.53[-4.60, +11.77]
Kev-4B	85	88.24	84.71	84.71	+0.00[-6.67, +6.10]
Kev-9B	85	83.53	88.24	84.71	-3.53[-9.88, +2.22]
NanoJev	85	40.00	41.18	38.82	-2.35[-9.64, +4.71]
Laya	85	64.71	61.18	65.88	+4.71[-4.76, +14.29]

Table 3: Fixed-profile external JevBench authored cohort: 85 tasks in 49 clusters, with 98 long hard tasks excluded by the declared screen. No profile, temperature, or quantizer fitting uses this cohort. Accuracy is %; differences are pp.

Model	Q. %	Native	Packed	Ratio %
Kev-0.8B	66.09	1437.1	741.1	51.6
Kev-4B	84.84	8026.8	3031.0	37.8
Kev-9B	87.15	15146.0	5460.1	36.0
NanoJev	73.86	2274.5	818.7	36.0
Laya	81.45	1607.1	472.6	29.4

Table 4: Parameter storage estimates in MiB. Packed includes retained native parameters and scales; Q. % is quantized parameter coverage. This is not a device-memory measurement.

Model	n	Dense	Packed	Peak
Kev-0.8B	41	1469.2	760.5	881.0
Kev-4B	41	8131.0	3123.0	3503.8
Kev-9B	41	15154.2	5488.1	6107.1
NanoJev	32	2284.4	832.2	946.8
Laya	41	1616.5	495.1	551.8

Table 5: Measured allocated CUDA memory (MiB) after conversion: dense/packed persistent allocation and packed inference peak. n counts scored parity decisions. Each packed path has zero measured JS and decision flips against the same artifact’s dense reconstruction on this small subset.

remains possible. The shared screen does not test long contexts or large candidate sets. NanoJev’s shooting reference-policy labels do not measure optimal behavior, human preferences, or successful outcomes.

Packed artifacts reduce storage but require native components and an initial dense load. Native integer kernels, direct packed-only loading, and fair dense/packed latency and throughput comparisons remain system work. These omissions bound the present claims rather than being substituted with projected performance.

8 Conclusion

S1Q provides a specified, reproducible PTQ adaptation for five open typed-decision checkpoints. It combines activation-aware scaling and clipping with optional categorical-Fisher probes and a development decision objective. Real packed weights and device measurements demonstrate reduced storage and persistent allocation, while probability and transfer results expose nonuniform quality changes. The evidence supports a cross-architecture empirical framework, not universal superiority, lossless W4, or native INT4 acceleration. The release enables stronger controlled baselines, broader frozen decision cohorts, and efficient kernels to be evaluated against a documented starting point.

Data and artifact availability. Source, pins, summaries, paired derived predictions, and recipes are public at github.com/CYMCharming/S1Q. Release v0.1.0 provides evaluated packed linears for Kev-0.8B, Kev-4B, Kev-9B, and Laya with byte checksums. NanoJev’s source license and fine-tuned weight redistribution grant are distinct; the audited card lacks a separate explicit weight grant, so recipes/results are released without mirroring packed weights. Raw dataset records remain local under upstream terms.

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A Probe Covariance Proof and Approximation Details

Let $D_q = \text{diag}(p_q)$ and $B_q = I - p_q \mathbf{1}^\top$. Then $v_q = B_q D_q^{1/2} \epsilon_q$ and

$$\mathbb{E}[v_q v_q^\top] = B_q D_q B_q^\top \quad (22)$$

$$= D_q - p_q p_q^\top. \quad (23)$$

Furthermore $\mathbf{1}^\top B_q = 0$. Independent zero-mean projections eliminate cross-question covariance terms in $g = Q^{-1/2} \sum_q J_q^\top v_q$, proving Equation (8). Using raw-logit Jacobians instead introduces an explicit T^{-2} factor; scaled-logit Jacobians already include it.

The squared-gradient statistic is unbiased conditional on the teacher and fixed records for the specified row average. Finite probes can have high variance. The collector’s denominator counts all flattened output rows across successful probes, including padded rows with zero gradients. Longer records contribute more rows. The experiment path uses raw moments followed by mean normalization and the floor in Equation (10); an optional shrinkage helper is not used in these runs.

The local objective drops off-diagonal entries of the propagated Fisher and input second-moment matrix, token-position correlations, joint activation–sensitivity dependence, and cross-layer effects. It evaluates reconstructed weights in FP32, whereas the native execution rounds transformed weights and compensated inputs in their actual compute dtypes. These assumptions limit any interpretation of the objective as exact end-to-end KL minimization.

B Full Development Search

The selection grid does not include learned rotations, differentiable rounding, full Hessian compensation, or supervised-gradient alternatives to the categorical probes. Fisher and activation collection use the pristine model rather than a partially quantized model. Different profiles restore original weights before recomputation. Protection is based on the RTN relative-error score, not on a test outcome.

Temperature scaling is fitted only after selection using positive per-type temperatures and separate labeled data. Raw logits are used for development. The native reference follows this raw-logit convention rather than an upstream served temperature or rounded response.

C Main Probability Metrics

NLL is $-\log(\max(p_y, 10^{-15}))$. Multiclass Brier is $\sum_k (p_k - \mathbf{1}[k = y])^2$, including both candidates in Boolean questions. ECE uses 15 equal-width top-confidence bins. Expected-level MAE for a K -level score question is $|\sum_{k=0}^{K-1} k p_k - y|$, not the error of the argmax level. All

completed methods/cohorts are retained in the public metrics report.

D Source Revisions and Data Semantics

The final decision-v7 test sources are Amazon Reviews, TREC, BoolQ, MNLI, Yelp, IMDb, SST-5, AG News, and deterministic quantity-limit, return-window, spend-threshold, and age-eligibility rules. Banking77 and DBpedia14 appear upstream but do not enter this screened final test. Transfer-v4 includes Emotion, PAWS, QNLI, MMLU, Tweet Offensive, SciQ, and synthetic combination, exception, negation, authorization, and deadline tasks. These are converted subsets, not full original-benchmark scores or production decision logs.

NanoJev test includes 321 Basic and 702 Predict Position decisions; OOD contains 398 and 626. All supplied gold actions match recorded upstream reference-policy argmax targets. The reference policies are `edbeechning/doom_basic_1111` and `thainv0212/sonic_doom`. The converter preserves explicit labels. Neither maze/snake coverage nor human-labeled optimal action is implied by these cohorts.

Episode grouping joins repeated public requests transitively within upstream partitions. Admission reserves complete groups in the fixed priority test, OOD, development, calibration, removing lower-priority overlapping components before sampling. Original repeated observations are not silently deduplicated or reassigned into new test splits. Group hashing divides calibration into quantizer and temperature pools.

The external short cohort contains no admitted mathematical-probability items, even though ten such hard items exist in the full public source. It does not test recovery of known probability distributions. Original paraphrase pairs are kept in scenario clusters. Unknown foundation-model training overlap is not ruled out.

E Runtime and Reproducibility

A supported installation is checked separately in an isolated Python 3.12 CPU environment. Native imports, dependency consistency, 129 tests, and 28 subtests passed; CPU tests do not substitute for GPU evaluation. Some recorded GPU packages lie outside current upstream declarations. Replaying their exact versions differs from installing a newly supported environment and can affect outputs. Full runtime notes accompany the repository.

Summary and paired-prediction hashes bind the report to completed runs. Tables are derived from that report and saved development candidates. This draft uses experiment release v0.1.0 and code commit `c46a95b2438cb4c878e789ea3c538080d6df8da5`. The aggregate SHA-256 is

`8c75a1ecd0e3a8698b1773006c907096d95ab350f8e9dc9523a6665beb294d3d`

Profile	Kev-0.8B	Kev-4B	Kev-9B	NanoJev	Laya
RTN	77.81	85.53	84.57	81.96	65.59
Local	80.39	84.89	85.85	81.96	67.52
Local + protect	79.74	84.89	86.82	81.96	65.92
Local + A8 QDQ	80.39	84.57	85.85	81.96	67.20
Local + A4 QDQ	61.41	53.70	39.87	80.78	53.38
Fisher	81.67	85.21	86.17	82.35	64.31
Fisher + protect	81.03	85.21	86.17	81.96	64.95

Table 6: Recorded development accuracy (%) for every completed expanded-search profile. Shared development has 311 questions; NanoJev has 255 shooting reference-policy decisions. Accuracy is reported, but is not the profile-selection objective.

Profile	Kev-0.8B	Kev-4B	Kev-9B	NanoJev	Laya
RTN	0.02353	0.01049	0.01495	0.00213	0.03182
Local	0.02195	0.00555	0.00773	0.00068	0.03192
Local + protect	0.02214	0.00591	0.00797	0.00069	0.03354
Local + A8 QDQ	0.02486	0.00586	0.00617	0.00069	0.03310
Local + A4 QDQ	0.16023	0.23761	0.32635	0.01257	0.13100
Fisher	0.02089	0.00757	0.00704	0.00074	0.04633
Fisher + protect	0.02113	0.00743	0.00730	0.00111	0.04278

Table 7: Recorded development objective in Equation (20); lower is preferred. RTN is only a baseline. Optional protection retains 5% of eligible layer count, not 5% of parameters. All nonselected profiles remain visible rather than being omitted after testing.

The manuscript provenance file records hashes of every table input. Failed attempts, including a NanoJev out-of-memory attempt followed by an unchanged rerun on an idle GPU, remain documented.

Each exported packed file has a whole-file checksum; large files are split below the release-host limit with independently verified parts. Public downloading of Laya was tested end to end. Original code and upstream modifications retain applicable licenses and attribution. S1Q’s source license does not relicense model or data.

F Experiments Needed for Stronger Claims

This manuscript reports completed work rather than placeholder measurements. A stronger method-comparison submission requires official AWQ/GPTQ and, where compatible, OmniQuant under identical native interfaces and budgets. Uniform, categorical-Fisher, supervised-gradient, and uncertainty-selected calibration variants would distinguish estimator effects from search and data selection. Multiple seeds, probe counts, calibration sizes, and collector time/memory measurements are needed to quantify robustness and cost. Larger independently frozen datasets should extend context length, candidate count, label provenance, and architecture coverage. Direct packed loading and hardware kernels require paired dense latency/throughput tests before an acceleration claim. None is presented as a completed result in this draft.

Model	Method	NLL		Brier		ECE15	
		Raw	Temp.	Raw	Temp.	Raw	Temp.
Kev-0.8B	Native	0.631	0.475	0.276	0.253	0.107	0.035
	RTN	0.642	0.527	0.302	0.283	0.119	0.025
	S1Q	0.684	0.500	0.295	0.271	0.121	0.029
Kev-4B	Native	0.530	0.374	0.226	0.199	0.091	0.034
	RTN	0.544	0.403	0.231	0.212	0.088	0.020
	S1Q	0.560	0.390	0.241	0.211	0.098	0.026
Kev-9B	Native	0.500	0.369	0.210	0.192	0.079	0.039
	RTN	0.533	0.409	0.230	0.214	0.091	0.030
	S1Q	0.535	0.383	0.221	0.203	0.080	0.017
NanoJev	Native	0.547	0.534	0.279	0.282	0.077	0.051
	RTN	0.561	0.537	0.283	0.284	0.060	0.043
	S1Q	0.543	0.532	0.278	0.281	0.057	0.060
Laya	Native	1.442	0.843	0.501	0.461	0.177	0.099
	RTN	1.508	0.911	0.527	0.498	0.184	0.113
	S1Q	1.515	0.880	0.523	0.482	0.180	0.094

Table 8: Main-cohort probability metrics. Each method fits its own positive per-type temperature on the same separate partition; temperature does not change accuracy. RTN is precision-matched, including A8 QDQ for Kev-9B. These use dataset target labels, not measurements of real-world success frequencies.

Component	Repository	Immutable revision
Kev source	jaredpalmer/kev	0fe8fc97c2bcc247fa3efb6e5c32af4e99770e91
Kev-0.8B	jaredpalmer/kev-0.8b	9a45d25eb2ab761841196625383fa1dfff0e56c1e
Kev-4B	jaredpalmer/kev-4b	139fdd94f1b6a6ad80cc15e08fcb99cac885a101
Kev-9B	jaredpalmer/kev-9b	2629c06a5aeb0feb3b9783bafed17ed8f39ecf5c
NanoJev source	TianyuCodings/NanoJev	76fd9c9ecdca45a9bcef17991a07d3041a87685a
NanoJev checkpoint	C-Tianyu/NanoJev	047b927b30882a1138fc504821b82ac145a4b81a
NanoJev dataset	C-Tianyu/NanoJev-Data	7afc5257c0f3ff0ba08512729888a51d94b40e7e
Laya source	NandhaKishorM/laya	6d942c92081fbc139e736bbd9ac0023223c29b7f
Laya checkpoint	convaiinnovations/laya	55cf4c4ebb4e31b2550e8bdf3bd21b99753851
JevBench	fstandhartinger/jevbench	bb05a335bc809e61b20c0f745d25499a82b326fc

Table 9: Pinned sources, released checkpoints, and external data. Base-model/tokenizer pins and raw-file SHA-256 values are additionally recorded in the release configuration and run manifests.

Environment	GPU	PyTorch	Transformers	PEFT	NumPy
Kev recorded	A100 80GB	2.7.1+cu118	5.16.1	0.20.0	2.5.2
Nano/Laya recorded	A800 80GB	2.9.1+cu126	5.17.0	0.21.0	1.26.4

Table 10: Recorded experiment environments, not a claim that all versions meet the latest upstream declared dependency ranges. Exact run manifests take precedence.